

MAT1002 Lecture 4,

Outline

- Alternating series approximation (10.6)
- Conditional convergence and rearrangement
- Power series (10.7)

Alternating Series Approximation

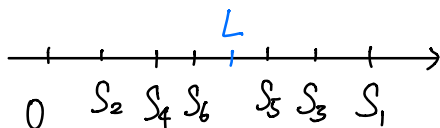
Suppose for a given series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$, we have

- $u_n > 0$, $\forall n \geq 1$;
- $\{u_n\}$ is nonincreasing;
- $u_n \rightarrow 0$ as $n \rightarrow \infty$.

Then $\sum_{n=1}^{\infty} (-1)^{n+1} u_n = L$ for some $L \in \mathbb{R}$. If we want to approximate L using the sum of the first K terms (S_K), then

- L is between S_K and S_{K+1} ;
- $\underbrace{|L - S_K|}_{\text{error}} < \underbrace{u_{K+1}}_{\text{first unused term, in absolute value}}$;
- $L - S_K$ has same sign as first unused term.

Intuition:



- $|L - S_5| = S_5 - L < S_5 - S_6 = u_6$
- First unused term: $-u_6$ (< 0)

e.g. Find the value of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(n-1)!}$, rounded to three decimal places.

Sol: • $u_n = \frac{1}{(n-1)!}$ is positive, nonincreasing;

• For $n \geq 2$, $0 < \frac{1}{(n-1)!} \leq \frac{1}{n-1} \rightarrow 0$,

So $\lim_{n \rightarrow \infty} u_n = 0$

- Hence series converges (by alternating series test).
- Try to look for a term $\frac{1}{(n-1)!}$ that is small, say < 0.001 ($= \frac{1}{1000}$).
- When $K=8$, $\frac{1}{(K-1)!} = \frac{1}{7!} = \frac{1}{5040} < \frac{1}{5000} = 0.0002$.
- Note that $S_7 = \frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{1}{6!}$
 $= 0.36805\dots$
- Since first unused term is $\frac{1}{7!}$, which is < 0.0002 , the first three decimal places of the series are given by 0.368.

Fact: exact value is $0.367879\dots \approx 0.368$.

Conditional Convergence

Def: A series $\sum a_n$ is said to be **convergent conditionally** if $\sum a_n$ converges but $\sum |a_n|$ diverges.

e.g. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^p}$ {

converges absolutely, if $p > 1$;
converges conditionally, if $0 < p \leq 1$;
diverges, if $p \leq 0$

↑
Alternating p-series

What is the key differences between absolute convergence and conditional convergence?

Re-arrangements (optional)

Consider the series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$, which converges (conditionally). Let

$$L := \sum_{n=3}^{\infty} (-1)^{n+1} \frac{1}{n} = \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots$$

Since $\frac{1}{3} - \frac{1}{4}$, $\frac{1}{5} - \frac{1}{6}$, $\dots$, $\frac{1}{2k-1} - \frac{1}{2k}$, $\dots$ are positive, $L > 0$.

Consider rearranging the terms in $\sum_{n=3}^{\infty} (-1)^{n+1} \frac{1}{n}$ as follows.

$$\frac{1}{3} - \frac{1}{4} - \frac{1}{6} + \frac{1}{5} - \frac{1}{8} - \frac{1}{10} + \frac{1}{7} - \frac{1}{12} - \frac{1}{14} + \dots$$

(One odd term followed by two even terms)

Note that $\frac{1}{3} - \frac{1}{4} - \frac{1}{6} = \frac{4-3-2}{12} < 0$, $\frac{1}{5} - \frac{1}{8} - \frac{1}{10} = \frac{8-5-4}{40} < 0$,

$$\dots, \frac{1}{2k+1} - \frac{1}{4k} - \frac{1}{4k+2} = \frac{8k-4k-2-4k}{16k^2+8k} < 0, \text{ so}$$

the rearranged series cannot converge to L , since $L > 0$.

Fact: Changing the order of the terms of a conditionally convergent series may change its value. In fact, a conditionally convergent series can converge to ANY value by rearranging its terms.

In contrast, an absolutely convergent series can only converge to one value, no matter how you order its terms.

Fact: If $\sum_{n=1}^{\infty} a_n$ converges absolutely, and $\sigma: \mathbb{Z}_+ \rightarrow \mathbb{Z}_+$ is any bijection (i.e., permutation of positive integers), then

$$\sum_{n=1}^{\infty} a_{\sigma(n)} = \sum_{n=1}^{\infty} a_n.$$

We omit the proofs of the facts.

Power Series

We now shift our attention to series of functions. Consider the

$$f(x) := 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n.$$

By geometric series,

$$f(x) = \begin{cases} \frac{1}{1-x}, & \text{if } |x| < 1; \\ \text{D.N.E.} & \text{if } |x| \geq 1. \\ \text{(divergent)} \end{cases}$$

In other words, whether $f(x)$ is defined depends on the value of x (just like other functions).

Definition

A **power series about $x = a$** is a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

$\{c_n\}$ sequence of real numbers

The number a above is called the **center** of the power series. In particular, a power series about $x = 0$ is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \dots$$

Remark: A power series can be thought of as a polynomial of infinite degree, but it may not converge for all $x \in \mathbb{R}$.

Intuition • For the power series $\sum_{n=0}^{\infty} x^n$, its partial sums are

$$S_0(x) = 1, S_1(x) = 1+x, \dots, S_n(x) = \sum_{k=0}^n x^k$$

Graph them using Desmos and observe the pattern.

• Do the same thing for $\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n (x-2)^n$. Note that

$$\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n (x-2)^n = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2}x\right)^n = \frac{1}{1 - 1 + \frac{1}{2}x} = \frac{2}{x},$$

given that $\left|1 - \frac{1}{2}x\right| < 1$, i.e., $\underbrace{0 < x < 4}$.

$(0, 4)$ is an interval centered at 2

Remark: • In power series notation, $0^0 := 1$, e.g. if

$$f(x) := \sum_{n=0}^{\infty} x^n, \text{ then } f(0) = \sum_{n=0}^{\infty} 0^n = 0^0 + 0^1 + 0^2 + \dots = 1.$$

Indeed, $f(x) = \frac{1}{1-x}$ for $|x| < 1$, so $f(0) = 1$.

• Any power series $\sum_{n=0}^{\infty} C_n(x-a)^n$ always converges at its center a , since

$$\sum_{n=0}^{\infty} C_n(a-a)^n = C_0.$$

Q: How to determine convergence of power series?

A: Try Ratio Test / Root Test.

Example

For what values of x do the following power series converge?

(a) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$. *Ans: Converges $\Leftrightarrow x \in (-1, 1]$.*

(b) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$. *Ans: Converges $\Leftrightarrow x \in (-\infty, \infty)$.*

(c) $\sum_{n=0}^{\infty} n! x^n$. *Ans: Converges $\Leftrightarrow x = 0$.*

(a) $\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x|^{n+1}}{n+1} \cdot \frac{n}{|x|^n} = |x| \cdot \frac{n}{n+1} \rightarrow |x| \text{ as } n \rightarrow \infty$

By Ratio Test, if $|x| < 1$, series converges (absolutely); if $|x| > 1$, series diverge. If $x = 1$, series converges (alt. harmonic); if $x = -1$, series diverges (negative harmonic).

(b) $\left| \frac{a_{n+1}}{a_n} \right| = \frac{|x|^{n+1}}{(n+1)!} \cdot \frac{n!}{|x|^n} = \frac{|x|}{n+1} \rightarrow 0 \text{ as } n \rightarrow \infty$

By Ratio Test, series always converges.

$$(c) \left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)! |x|^{n+1}}{n! |x|^n} = (n+1)|x| \rightarrow \begin{cases} 0, & \text{if } x=0; \\ \infty, & \text{if } x \neq 0. \end{cases}$$

By Ratio Test, series ONLY converges for $x=0$.

In the examples above, the convergence values of x form an interval (if we view $\{a\}$ as $[a, a]$). What is the general pattern?

Theorem If $\sum_{n=0}^{\infty} C_n(x-a)^n$ converges at $x=x_0$, then it converges absolutely for all x with $|x-a| < |x_0-a|$. If it diverges at $x=x_0$, then it diverges for all x with $|x-a| > |x_0-a|$.

Proof: Suppose series converges at $x=x_0$. May assume $x_0 \neq a$.

So $|x_0-a| > 0$. Since $\lim_{n \rightarrow \infty} C_n(x_0-a)^n = 0$, $\exists N$ s.t. $\forall n \geq N$,

$$|C_n(x_0-a)^n| < 1 \Rightarrow |C_n| < \frac{1}{|x_0-a|^n}. \text{ For any } x$$

with $|x-a| < |x_0-a|$, we have

$$0 \leq |C_n| |x-a|^n < \frac{|x-a|^n}{|x_0-a|^n} = \left| \frac{x-a}{x_0-a} \right|^n.$$

Since $\sum_{n=0}^{\infty} \left| \frac{x-a}{x_0-a} \right|^n$ converges, $\sum_{n=0}^{\infty} |C_n| |x-a|^n$ converges, so

$\sum C_n(x-a)^n$ converges absolutely.

For the second part, suppose $\sum C_n(x_0-a)^n$ diverges. If $\exists x$ s.t. $|x-a| > |x_0-a|$ but $\sum C_n(x-a)^n$ converges, then by the first part, $\sum C_n|x_0-a|^n$ would converge (absolutely). This is a contradiction, so no such x can exist, i.e. if $|x-a| > |x_0-a|$ then $\sum C_n(x-a)^n$ diverges. □

Radius of Convergence

Theorem (Existence of the Radius of Convergence)

For any power series $\sum c_n(x-a)^n$, one of the following three statements holds:

- (i) There exists a positive real number R such that the series converges absolutely for all x with $|x-a| < R$ but diverges for all x with $|x-a| > R$. The series may or may not converge for x with $|x-a| = R$.
- (ii) The series converges absolutely for all x ($R = \infty$).
- (iii) The series converges at $x = a$ and diverges elsewhere ($R = 0$).

The number R above is called the **radius of convergence** of the power series.